🏆 Infosys SE Elite Crack Vault — 12 College ITM Quantitative Aptitude 70Q Exam Compendium Elite

🎯 Infosys Systems Engineer (SE) | 📊 49 Questions | 🗓️ 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

Problem 1 • Infosys SE Question 1

MUST-SOLVE

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q16)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
In how many ways can the letters of COMMITTEE be arranged so that no two vowels are adjacent?

A	5400	B	3600
C	10800	D	2700

✓ **Verified Answer:** (A)** — `5400` **Detailed Step-by-Step Solution & Mathematical Derivation:** Word COMMITTEE contains 9 letters: C(1), O(1), M(2), I(1), T(2), E(2). Vowels: O, I, E, E (4 vowels, with two E's). Consonants: C, M, M, T, T (5 consonants, with two M's and two T's). 1) Arrange the 5 consonants first: Number of ways = 5! / (2! × 2!) = 120 / 4 = 30 ways. 2) 5 consonants create 6 available gaps: _ C _ M _ M _ T _ T _ To ensure no two vowels are adjacent, choose 4 gaps out of 6 for the 4 vowels: C(6, 4) = 15 ways. 3) Arrange the 4 vowels (O, I, E, E) in the chosen 4 gaps: Number of ways = 4! / 2! = 24 / 2 = 12 ways. Total arrangements = 30 × 15 × 12 = 5,400.

Problem 2 • Infosys SE Question 2

MUST-SOLVE

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q17)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
A committee of 5 is to be formed from 5 boys and 4 girls with at least 2 boys and at least 2 girls. In how many ways can this be done?

A	90	B	100
C	105	D	120

✓ **Verified Answer:** (B)** — `100` **Detailed Step-by-Step Solution & Mathematical Derivation:** Committee of 5 from 5 boys and 4 girls with at least 2 boys and at least 2 girls. Possible compositions (Boys, Girls): - Case 1: 2 Boys and 3 Girls => C(5, 2) × C(4, 3) = 10 × 4 = 40 ways. - Case 2: 3 Boys and 2 Girls => C(5, 3) × C(4, 2) = 10 × 6 = 60 ways. Total possible ways = 40 + 60 = 100 ways.

Problem 3 • Infosys SE Question 3

MUST-SOLVE

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q18)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
Two fair dice are thrown. Given that at least one die shows 4, what is the probability that the sum is a prime number?

A	4/11	B	1/3
C	5/11	D	4/9

✓ **Verified Answer:** (A)** — `4/11` **Detailed Step-by-Step Solution & Mathematical Derivation:** Two fair dice thrown. Total outcomes with at least one 4: Sample space S = {(4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (1,4), (2,4), (3,4), (5,4), (6,4)} => n(S) = 11 outcomes. Sums for each outcome: - (4,1) -> 5 (prime) - (4,2) -> 6 - (4,3) -> 7 (prime) - (4,4) -> 8 - (4,5) -> 9 - (4,6) -> 10 - (1,4) -> 5 (prime) - (2,4) -> 6 - (3,4) -> 7 (prime) - (5,4) -> 9 - (6,4) -> 10 Favorable outcomes where sum is prime: {(4,1), (4,3), (1,4), (3,4)} => 4 outcomes. Probability = 4 / 11.

Problem 4 • Infosys SE Question 4

MUST-SOLVE

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q19)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
A bag has 5 red, 4 blue and 3 green balls. Three balls are drawn at random without replacement. What is the probability that all three are of different colours?

A	3/11	B	1/4
C	2/11	D	5/22

✓ **Verified Answer:** (A)** — `3/11` **Detailed Step-by-Step Solution & Mathematical Derivation:** Total balls = 5 red + 4 blue + 3 green = 12 balls. 3 balls drawn at random without replacement. Total ways = C(12, 3) = (12 × 11 × 10) / (3 × 2 × 1) = 220. Favorable ways to get one of each colour = C(5, 1) × C(4, 1) × C(3, 1) = 5 × 4 × 3 = 60. Probability = 60 / 220 = 3 / 11.

Problem 5 • Infosys SE Question 5

MUST-SOLVE

ID: Q5 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q20)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
A shopkeeper marks his goods 40% above cost price and allows two successive discounts of 10% and 15%. His net profit percentage is:

A 7.1%	**B** 5.5%
C 8.6%	**D** 6.4%

✓ **Verified Answer:** (A)** — `7.1%` **Detailed Step-by-Step Solution & Mathematical Derivation:** Let Cost Price (CP) = 100. Marked Price (MP) = 100 × 1.40 = 140. Successive discounts of 10% and 15%: Selling Price (SP) = 140 × (1 - 0.10) × (1 - 0.15) = 140 × 0.90 × 0.85 = 126 × 0.85 = 107.1. Net profit = 107.1 - 100 = 7.1%.

Problem 6 • Infosys SE Question 6

MUST-SOLVE

ID: Q6 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q21)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_01_page_03.png

Question Text:
A, B and C can individually finish a job in 12, 18 and 24 days respectively. They work on alternate days in the order A, B, C, A, B, C, ... (one person per day, starting with A). On which day is the job completed?

A 16th	**B** 17th
C 18th	**D** 19th

✓ **Verified Answer:** (B)** — `17th` **Detailed Step-by-Step Solution & Mathematical Derivation:** Let total work = LCM(12, 18, 24) = 72 units. Daily efficiencies: - A = 72 / 12 = 6 units/day - B = 72 / 18 = 4 units/day - C = 72 / 24 = 3 units/day They work on alternate days: A on Day 1, B on Day 2, C on Day 3, A on Day 4, etc. In 1 cycle of 3 days, work done = 6 + 4 + 3 = 13 units. In 5 full cycles (15 days), work done = 5 × 13 = 65 units. Remaining work = 72 - 65 = 7 units. - Day 16: A works and completes 6 units. Total work = 65 + 6 = 71 units. Remaining = 1 unit. - Day 17: B works. Since B does 4 units/day, B finishes the remaining 1 unit in 1/4 of a day. Therefore, the job is completed on the 17th day.

Problem 7 • Infosys SE Question 7

MUST-SOLVE

ID: Q7 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q22)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_01_page_03.png

Question Text:
Two trains of lengths 240 m and 160 m run on parallel tracks. Running in opposite directions they cross each other in 8 s; running in the same direction the faster overtakes the slower in 40 s. The speed of the faster train is:

A 90 km/h	**B** 108 km/h
C 120 km/h	**D** 72 km/h

✓ **Verified Answer:** (B)** — `108 km/h` **Detailed Step-by-Step Solution & Mathematical Derivation:** Total distance to clear = 240 m + 160 m = 400 m. Let speed of faster train = v1 m/s and slower train = v2 m/s. 1) Running in opposite directions, relative speed = v1 + v2 = 400 / 8 = 50 m/s. 2) Running in same direction, relative speed = v1 - v2 = 400 / 40 = 10 m/s. Adding the two equations: 2·v1 = 60 => v1 = 30 m/s. Converting to km/h: 30 × (18 / 5) = 108 km/h.

Problem 8 • Infosys SE Question 8

MUST-SOLVE

ID: Q8 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q23)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_01_page_03.png

Question Text:
In an election, 20,000 voters were registered. 80% voted, and 5% of the votes cast were invalid. The winner got 55% of the valid votes. By how many votes did he win?

A 1520	**B** 1600
C 1360	**D** 1440

✓ **Verified Answer:** (A)** — `1520` **Detailed Step-by-Step Solution & Mathematical Derivation:** Total registered voters = 20,000. Voters who voted = 80% of 20,000 = 16,000. Invalid votes = 5% of 16,000 = 800. Valid votes = 16,000 - 800 = 15,200. Winner obtained 55% of valid votes => Loser obtained (100% - 55%) = 45%. Winning margin = 55% - 45% = 10% of valid votes. Winning margin = 0.10 × 15,200 = 1,520 votes.

Problem 9 • Infosys SE Question 9

MUST-SOLVE

ID: Q9 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q24)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_01_page_03.png

Question Text:
The difference between compound interest (annual) and simple interest on a sum for 3 years at 10% p.a. is ₹310. Find the sum.

A ₹8,000	B ₹9,000
C ₹10,000	D ₹12,000

✓ **Verified Answer:** (C)** — `₹10,000` **Detailed Step-by-Step Solution & Mathematical Derivation:** Difference between CI and SI for 3 years at rate r%: Formula: Difference = P × (r / 100)² × (3 + r / 100). Given r = 10% and Difference = ₹310: 310 = P × (0.10)² × (3 + 0.10) 310 = P × 0.01 × 3.10 = P × 0.031 P = 310 / 0.031 = ₹10,000.

Problem 10 • Infosys SE Question 10

MUST-SOLVE

ID: Q10 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 1 - Q25)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_01_page_03.png

Question Text:
₹12,000 is invested partly at 8% simple interest and the rest at 10% compounded annually. After 2 years the total interest is ₹2,270. How much was invested at simple interest?

A ₹4,000	B ₹5,000
C ₹6,000	D ₹7,000

✓ **Verified Answer:** (B)** — `₹5,000` **Detailed Step-by-Step Solution & Mathematical Derivation:** Let sum invested at 8% SI = ₹x. Then sum invested at 10% CI = ₹(12,000 - x). Time = 2 years. - Simple Interest at 8% for 2 years = (x × 8 × 2) / 100 = 0.16x. - Compound Interest at 10% for 2 years = (12,000 - x) × [(1 + 0.10)² - 1] = (12,000 - x) × 0.21 = 2,520 - 0.21x. Total interest = 0.16x + 2,520 - 0.21x = 2,520 - 0.05x. Given total interest = ₹2,270: 2,520 - 0.05x = 2,270 0.05x = 250 x = 250 / 0.05 = ₹5,000.

Problem 11 • Infosys SE Question 11

MUST-SOLVE

ID: Q11 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q16)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
In how many ways can the letters of the word MANAGEMENT be arranged so that all the vowels come together?

A	7560	B	15120
C	3780	D	30240

✓ **Verified Answer:** (A)** — `7560` **Detailed Step-by-Step Solution & Mathematical Derivation:** MANAGEMENT has M2, A2, N2, G1, E2, T1 (10 letters). Vowels A,A,E,E form one block: internal arrangements = 4!/(2!2!) = 6. Treating the block as one unit alongside the 6 consonants (M2,N2,G1,T1) gives 7 units: 7!/(2!2!) = 1260. Total = 1260 × 6 = 7,560.

Problem 12 • Infosys SE Question 12

MUST-SOLVE

ID: Q12 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q17)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
From a group of 6 men and 5 women, a team of 4 is to be selected such that it contains at most 2 women. In how many ways can this be done?

A	265	B	295
C	310	D	325

✓ **Verified Answer:** (A)** — `265` **Detailed Step-by-Step Solution & Mathematical Derivation:** 0 women: C(6,4)=15. 1 woman: C(6,3)×C(5,1)=100. 2 women: C(6,2)×C(5,2)=150. Total = 15+100+150 = 265.

Problem 13 • Infosys SE Question 13

MUST-SOLVE

ID: Q13 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q18)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
Three coins are tossed together, and it is known that at least one head appeared. What is the probability that exactly two heads appeared?

A 3/7	**B** 1/2
C 3/8	**D** 2/7

✓ **Verified Answer:** (A)** — `3/7` **Detailed Step-by-Step Solution & Mathematical Derivation:** Excluding TTT leaves 7 equally likely outcomes. Exactly two heads: HHT, HTH, THH = 3 outcomes. Probability = 3/7.

Problem 14 • Infosys SE Question 14

MUST-SOLVE

ID: Q14 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q19)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
A box contains 6 defective and 14 non-defective bulbs. Two bulbs are drawn at random without replacement. What is the probability that at least one is defective?

A 99/190	**B** 91/190
C 33/95	**D** 29/95

✓ **Verified Answer:** (A)** — `99/190` **Detailed Step-by-Step Solution & Mathematical Derivation:** P(no defective) = (14×13)/(20×19) = 91/190. P(at least one defective) = 1 − 91/190 = 99/190.

Problem 15 • Infosys SE Question 15

MUST-SOLVE

ID: Q15 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q20)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
A trader marks his goods at a price that would give him a 25% profit, but he actually gives a discount of 12% on the marked price and also uses a weight that is 10% less than it should be. Find his actual profit percentage.

A 20%	**B** 22.2%
C 25%	**D** 18%

✓ **Verified Answer:** (B)** — `22.2%` **Detailed Step-by-Step Solution & Mathematical Derivation:** Take CP = 100 per unit weight, so marked price = 125 (25% intended profit). After a 12% discount, SP = 125 × 0.88 = 110. But he delivers only 0.9 units (10% short weight) whose true cost is 90. Profit = 110 − 90 = 20 on a cost of 90, giving 20/90 × 100 ≈ 22.2%.

Problem 16 • Infosys SE Question 16

MUST-SOLVE

ID: Q16 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q21)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
A and B together can complete a task in 15 days. They worked together for 5 days, after which A left and B completed the remaining work alone in 20 more days. In how many days can A alone complete the task?

A 20 days	**B** 24 days
C 25 days	**D** 30 days

✓ **Verified Answer:** (D)** — `30 days` **Detailed Step-by-Step Solution & Mathematical Derivation:** In 5 days together, they finish 5/15 = 1/3. The remaining 2/3 is done by B alone in 20 days, so B's rate = (2/3)/20 = 1/30/day (B alone takes 30 days). Combined rate = 1/15, so A's rate = 1/15 − 1/30 = 1/30/day — A alone also takes 30 days.

Problem 17 • Infosys SE Question 17

MUST-SOLVE

ID: Q17 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q22)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
A boat's speed in still water is 15 km/h. It travels 60 km downstream and returns upstream to the starting point, taking a total of 9 hours. Find the speed of the stream.

A 4 km/h	**B** 5 km/h
C 6 km/h	**D** 3 km/h

✓ **Verified Answer:** (B)** — `5 km/h` **Detailed Step-by-Step Solution & Mathematical Derivation:** 60/(15+s) + 60/(15−s) = 9. Simplifying: 1800 = 9(225 − s^2), so s^2 = 25, s = 5 km/h.

Problem 18 • Infosys SE Question 18

CORE SE

ID: Q18 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q23)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_02_page_03.png

Question Text:
A town's population increases by 10% in the first year and decreases by 8% in the second year. If the population at the end of two years is 27,324, what was the population at the start?

A 27,000	**B** 26,500
C 26,000	**D** 27,500

✓ **Verified Answer:** (A)** — `27,000` **Detailed Step-by-Step Solution & Mathematical Derivation:** P × 1.10 × 0.92 = 27,324, so P × 1.012 = 27,324, giving P = 27,000.

Problem 19 • Infosys SE Question 19

CORE SE

ID: Q19 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q24)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_02_page_03.png

Question Text:
A sum of money amounts to ₹13,000 after 3 years and to ₹15,000 after 5 years at the same rate of simple interest. Find the rate of interest per annum.

A 8%	**B** 10%
C 12%	**D** 9%

✓ **Verified Answer:** (B)** — `10%` **Detailed Step-by-Step Solution & Mathematical Derivation:** Interest for 2 years (year 5 − year 3) = 15,000 − 13,000 = 2,000, so 1 year's interest = 1,000. Principal = 13,000 − 3×1,000 = 10,000. Rate = 1,000/10,000 × 100 = 10%.

Problem 20 • Infosys SE Question 20

CORE SE

ID: Q20 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 2 - Q25)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_02_page_03.png

Question Text:
₹16,000 is invested for 2 years at compound interest, compounded annually, and yields the same interest as ₹18,000 invested for 2 years at 10% simple interest. Find the rate of compound interest (approximately).

A 9.5%	**B** 10.5%
C 11.5%	**D** 12.5%

✓ **Verified Answer:** (B)** — `10.5%` **Detailed Step-by-Step Solution & Mathematical Derivation:** SI on ₹18,000 at 10% for 2 years = ₹3,600. Setting CI on ₹16,000 equal: $16,000\times((1+r)^2 - 1) = 3,600$ gives $(1+r)^2 = 1.225$, so $r \approx 10.5\%$.

Problem 21 • Infosys SE Question 21

CORE SE

ID: Q21 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q16)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
In how many ways can 8 different books be distributed among 3 students so that one student gets 4 books and the other two get 2 books each?

A	1260	**B**	2520
C	630	**D**	420

✓ **Verified Answer:** (A)** — `1260` **Detailed Step-by-Step Solution & Mathematical Derivation:** Choose which of the 3 students gets 4 books: 3 ways. Choose that student's 4 books: C(8,4)=70. Split the remaining 4 books between the other two specific students, 2 each: C(4,2)=6. Total = 3×70×6 = 1,260.

Problem 22 • Infosys SE Question 22

CORE SE

ID: Q22 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q17)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
A committee of 6 is to be formed from 7 men and 6 women such that the committee has more men than women. In how many ways can this be done?

A	658	**B**	686
C	700	**D**	623

✓ **Verified Answer:** (A)** — `658` **Detailed Step-by-Step Solution & Mathematical Derivation:** Valid (men,women) splits with men>women, total=6: (6,0): C(7,6)×C(6,0)=7. (5,1): C(7,5)×C(6,1)=21×6=126. (4,2): C(7,4)×C(6,2)=35×15=525. Total = 7+126+525 = 658. (Cross-check: adding the equal split (3,3)=700 and women>men splits=358 gives 658+700+358=1,716=C(13,6), confirming the count.)

Problem 23 • Infosys SE Question 23

CORE SE

ID: Q23 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q18)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
Bag A has 4 red and 6 black balls; Bag B has 7 red and 3 black balls. A bag is chosen at random and one ball is drawn from it. Given that the ball drawn is red, what is the probability that it came from Bag A?

A 4/11	**B** 7/11
C 2/5	**D** 1/2

✓ **Verified Answer:** (A)** — `4/11` **Detailed Step-by-Step Solution & Mathematical Derivation:** P(Red|A)=4/10, P(Red|B)=7/10, each bag chosen with probability 1/2. P(Red) = ½×0.4 + ½×0.7 = 0.55. By Bayes' theorem, P(A|Red) = (½×0.4)/0.55 = 0.2/0.55 = 4/11.

Problem 24 • Infosys SE Question 24

CORE SE

ID: Q24 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q19)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
A drawer has 5 red pens, 4 blue pens and 3 black pens. Four pens are drawn at random. What is the probability that exactly 2 are red and 2 are blue?

A 4/33	**B** 3/33
C 5/33	**D** 4/29

✓ **Verified Answer:** (A)** — `4/33` **Detailed Step-by-Step Solution & Mathematical Derivation:** Favourable = C(5,2)×C(4,2) = 10×6 = 60. Total ways = C(12,4) = 495. Probability = 60/495 = 4/33.

Problem 25 • Infosys SE Question 25

CORE SE

ID: Q25 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q20)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
A dealer's cost price for a mixture of two rice varieties (mixed 3:2 by weight at ₹40/kg and ₹60/kg) works out to ₹48/kg. He marks up the price by 25% and sells it, but uses a false weight that gives only 900 g for every 1 kg paid for. Find his actual profit percentage.

A 38.9%	**B** 35%
C 41.7%	**D** 33.3%

✓ **Verified Answer:** (A)** — `38.9%` **Detailed Step-by-Step Solution & Mathematical Derivation:** Marked-up SP for a claimed 1 kg = 48×1.25 = ₹60. True cost of the 900 g actually given = 0.9×48 = ₹43.2. Profit = 60−43.2 = ₹16.8 on a true cost of ₹43.2, giving 16.8/43.2×100 ≈ 38.9%.

Problem 26 • Infosys SE Question 26

CORE SE

ID: Q26 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q21)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
Pipe A can fill a tank in 12 hours, Pipe B in 15 hours, and a leak can empty the full tank in 20 hours. All three are opened together for 4 hours, after which the leak is plugged. How much more time do A and B together need to fill the remaining tank?

A 4 hours	**B** 3 hours
C 5 hours	**D** 6 hours

✓ **Verified Answer:** (A)** — `4 hours` **Detailed Step-by-Step Solution & Mathematical Derivation:** Combined rate (A+B−leak) = 1/12+1/15−1/20 = 5/60+4/60−3/60 = 1/10 per hour. In 4 hours, 4/10 = 2/5 of the tank fills, leaving 3/5. A+B's combined rate (no leak) = 1/12+1/15 = 3/20 per hour. Time for remaining 3/5 = (3/5)÷(3/20) = 4 hours.

Problem 27 • Infosys SE Question 27

CORE SE

ID: Q27 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q22)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
In a 1000 m race, A beats B by 100 m, and in a separate 1000 m race, B beats C by 100 m (each runner maintains a constant speed across races). By how many metres does A beat C in a 1000 m race between them?

A 190 m	**B** 200 m
C 180 m	**D** 210 m

✓ **Verified Answer:** (A)** — `190 m` **Detailed Step-by-Step Solution & Mathematical Derivation:** Speed ratio A:B = 1000:900 = 10:9 (A beats B by 100 m in 1000 m). Similarly B:C = 10:9. So A:C = (10/9)×(10/9) = 100:81. When A covers 1000 m, C covers (81/100)×1000 = 810 m. A beats C by 1000−810 = 190 m.

Problem 28 • Infosys SE Question 28

CORE SE

ID: Q28 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q23)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
An alloy contains 60% copper and 40% zinc. 20 kg of this alloy is melted and mixed with x kg of pure copper to form a new alloy containing 75% copper. Find x.

A 12 kg	**B** 10 kg
C 15 kg	**D** 8 kg

✓ **Verified Answer:** (A)** — `12 kg` **Detailed Step-by-Step Solution & Mathematical Derivation:** Copper in original alloy = 0.6×20 = 12 kg; zinc = 8 kg (unchanged after adding pure copper). For 75% copper (25% zinc): 8 = 0.25×(20+x), so 20+x = 32, x = 12 kg.

Problem 29 • Infosys SE Question 29

CORE SE

ID: Q29 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q24)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
₹8,000 is invested for 1 year at 10% per annum, compounded half-yearly. Find the difference between the compound interest and the simple interest earned on the same sum, at the same rate, for the same period.

A ₹20	**B** ₹40
C ₹80	**D** ₹100

✓ **Verified Answer:** (A)** — `₹20` **Detailed Step-by-Step Solution & Mathematical Derivation:** Half-yearly compounding: 2 periods at 5% each. Amount = 8000×1.05×1.05 = 8,820, so CI = ₹820. SI = 8000×10%×1 = ₹800. Difference = 820−800 = ₹20.

Problem 30 • Infosys SE Question 30

CORE SE

ID: Q30 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 3 - Q25)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
In what time will ₹8,000 amount to ₹9,261 at 5% per annum, compounded annually?

A 2 years	**B** 3 years
C 4 years	**D** 3.5 years

✓ **Verified Answer:** (B)** — `3 years` **Detailed Step-by-Step Solution & Mathematical Derivation:** 9,261/8,000 = 1.157625 = 1.05^3 (since 1.05^2=1.1025, ×1.05=1.157625). So the time is 3 years.

Problem 31 • Infosys SE Question 31

CORE SE

ID: Q31 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q16)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
In how many ways can 6 men and 4 women be seated at a round table such that no two women sit together?

A	43,200	B	86,400
C	21,600	D	14,400

✓ **Verified Answer:** (A)** — `43,200` **Detailed Step-by-Step Solution & Mathematical Derivation:** Seat 6 men in a circle: (6−1)! = 120 ways. This creates 6 gaps; choose and arrange 4 women in 4 of these 6 distinct gaps: P(6,4) = 6×5×4×3 = 360. Total = 120×360 = 43,200.

Problem 32 • Infosys SE Question 32

CORE SE

ID: Q32 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q17)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
In how many ways can the letters of the word ORANGE be arranged so that the vowels occupy only the odd positions?

A	36	B	72
C	18	D	144

✓ **Verified Answer:** (A)** — `36` **Detailed Step-by-Step Solution & Mathematical Derivation:** ORANGE has 3 vowels (O,A,E) and 3 consonants (R,N,G), matching the 3 odd and 3 even positions exactly. Vowels in odd positions: 3! = 6. Consonants in even positions: 3! = 6. Total = 6×6 = 36.

Problem 33 • Infosys SE Question 33

CORE SE

ID: Q33 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q18)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
Three machines A, B and C produce 25%, 35% and 40% of a factory's output respectively, with defect rates of 5%, 4% and 2%. An item is picked at random and found defective. What is the probability it came from machine C?

A 16/69	**B** 8/69
C 23/69	**D** 25/69

✓ **Verified Answer:** (A)** — `16/69` **Detailed Step-by-Step Solution & Mathematical Derivation:** P(D) = 0.25×0.05 + 0.35×0.04 + 0.40×0.02 = 0.0125+0.014+0.008 = 0.0345. P(C|D) = (0.40×0.02)/0.0345 = 0.008/0.0345 = 16/69.

Problem 34 • Infosys SE Question 34

CORE SE

ID: Q34 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q19)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
5 boys and 3 girls are seated randomly in a row of 8 chairs. What is the probability that all 3 girls sit together?

A 3/28	**B** 1/8
C 3/56	**D** 5/28

✓ **Verified Answer:** (A)** — `3/28` **Detailed Step-by-Step Solution & Mathematical Derivation:** Treating the 3 girls as one block: favourable arrangements = 6!×3! = 720×6 = 4,320. Total arrangements = 8! = 40,320. Probability = 4320/40320 = 3/28.

Problem 35 • Infosys SE Question 35

CORE SE

ID: Q35 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q20)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
A trader buys 80 kg of rice at ₹25/kg. He sells 50 kg at a 20% profit and wants his overall profit on all 80 kg to be 15%. At what price per kg must he sell the remaining 30 kg?

A ₹26.67	**B** ₹25
C ₹28	**D** ₹24

✓ **Verified Answer:** (A)** — `₹26.67` **Detailed Step-by-Step Solution & Mathematical Derivation:** Total CP = 80×25 = ₹2,000; target total SP (15% profit) = ₹2,300. SP from the first 50 kg (cost ₹1,250, at 20% profit) = ₹1,500. Remaining SP needed = 2300−1500 = ₹800 for 30 kg → ₹800/30 ≈ ₹26.67/kg.

Problem 36 • Infosys SE Question 36

CORE SE

ID: Q36 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q21)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
A, B and C can complete a work alone in 20, 30 and 60 days respectively. All three start together. After 4 days, B leaves; after 3 more days, C also leaves, and A finishes the rest alone. In how many total days is the work completed?

A 15 days	**B** 14 days
C 16 days	**D** 13 days

✓ **Verified Answer:** (A)** — `15 days` **Detailed Step-by-Step Solution & Mathematical Derivation:** Rates (LCM 60): A=3/60, B=2/60, C=1/60 per day. First 4 days (all three): 4×6/60=24/60 done. Next 3 days (A+C only): 3×4/60=12/60 done — total 36/60=3/5 done, 2/5=24/60 remains. A alone at 3/60/day takes 24/60÷3/60 = 8 more days. Total = 4+3+8 = 15 days.

Problem 37 • Infosys SE Question 37

CORE SE

ID: Q37 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q22)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
Two trains, 300 km apart, move towards each other on the same track. Train A departs at 8:00 AM at 60 km/h; Train B departs at 8:30 AM at 90 km/h. At what time do they meet?

A	10:18 AM	B	10:12 AM
C	10:06 AM	D	10:24 AM

✓ **Verified Answer:** (A)** — `10:18 AM` **Detailed Step-by-Step Solution & Mathematical Derivation:** From 8:00 to 8:30, only A moves: 60×0.5=30 km, leaving 270 km. From 8:30, combined speed = 150 km/h; time = 270/150 = 1.8 hr = 1 hr 48 min. Meeting time = 8:30 AM + 1:48 = 10:18 AM.

Problem 38 • Infosys SE Question 38

CORE SE

ID: Q38 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q23)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
In a two-candidate election, the winner received 60% of votes cast and won by 4,800 votes. If 20% of registered voters did not vote and there were no invalid votes, how many voters were registered?

A	30,000	B	28,000
C	32,000	D	24,000

✓ **Verified Answer:** (A)** — `30,000` **Detailed Step-by-Step Solution & Mathematical Derivation:** Winner's margin (60% −40%=20% of votes cast) = 4,800 → votes cast = 4800/0.20 = 24,000. Since this is 80% of registered voters: registered = 24000/0.80 = 30,000.

Problem 39 • Infosys SE Question 39

CORE SE

ID: Q39 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q24)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
Find the compound interest on ₹12,000 for 1.5 years at 10% per annum, compounded annually (interest for the fractional half-year is simple interest on the amount at the end of the first year).

A	₹3,860	B	₹3,800
C	₹3,960	D	₹3,720

✓ **Verified Answer:** (A)** — `₹3,860` **Detailed Step-by-Step Solution & Mathematical Derivation:** Year 1: 12000×1.10 = 13,200 (CI so far = 1,200). For the next half-year, SI on 13,200 at 10% for 0.5 yr = 660. Total amount = 13,860; total CI = 13860−12000 = ₹3,860.

Problem 40 • Infosys SE Question 40

CORE SE

ID: Q40 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 4 - Q25)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
The compound interest on a sum for 2 years is ₹1,050, and for 3 years is ₹1,655, at the same annual rate. The simple interest on the same sum for 2 years, at the same rate, is ₹1,000. Find the rate of interest per annum.

A	10%	B	12%
C	8%	D	15%

✓ **Verified Answer:** (A)** — `10%` **Detailed Step-by-Step Solution & Mathematical Derivation:** 3rd-year interest = 1655−1050 = ₹605 — but more directly: $CI_2 - SI_2 = Pr^2 = 1050 - 1000 = 50$, and $SI_2 = 2Pr = 1000$ so Pr=500. Dividing, r = 50/500 = 0.10 = 10% (and P=₹5,000, verified: $5000 \times 1.1^3 = 6,655$, $CI_3 = 1,655$ ✓).

Problem 41 • Infosys SE Question 41

CORE SE

ID: Q41 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q16)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
A committee of 5 is to be formed from 6 men and 5 women. A specific man X must be included, a specific woman Y must be excluded, and the committee must include at least 1 woman. In how many ways can this be done?

A 121	**B** 126
C 120	**D** 115

✓ **Verified Answer:** (A)** — `121` **Detailed Step-by-Step Solution & Mathematical Derivation:** With X fixed in and Y fixed out, the remaining pool is 5 men + 4 women = 9 people, from which 4 more members are chosen: C(9,4)=126. Subtract the invalid case where all 4 additional members are men (no woman at all): C(5,4)=5. Valid = 126−5 = 121.

Problem 42 • Infosys SE Question 42

CORE SE

ID: Q42 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q17)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
In how many ways can the 8 distinct letters of COMPUTER be arranged such that C and T have exactly 2 letters between them (in either order)?

A 7200	**B** 6000
C 8400	**D** 5040

✓ **Verified Answer:** (A)** — `7200` **Detailed Step-by-Step Solution & Mathematical Derivation:** Valid position-pairs with C and T exactly 3 apart: 5 pairs with C before T, plus 5 with T before C = 10 total. The remaining 6 distinct letters fill the remaining 6 positions in 6!=720 ways. Total = 10×720 = 7,200.

Problem 43 • Infosys SE Question 43

CORE SE

ID: Q43 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q18)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
Two tickets are drawn one after another, without replacement, from a box containing tickets numbered 1 to 10. What is the probability that the number on the second ticket is greater than the number on the first?

A 1/2	**B** 9/20
C 11/20	**D** 2/5

✓ **Verified Answer:** (A)** — `1/2` **Detailed Step-by-Step Solution & Mathematical Derivation:** By symmetry, for any two distinct numbers drawn in some order, exactly half the time the second exceeds the first. Probability = 1/2 (confirmed via direct count: 45 favourable ordered outcomes out of 90).

Problem 44 • Infosys SE Question 44

CORE SE

ID: Q44 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q19)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
A mini-deck has 16 cards: 4 suits with 4 cards each. 4 cards are drawn at random. What is the probability that at least 2 of them share the same suit?

A 391/455	**B** 64/455
C 327/455	**D** 201/455

✓ **Verified Answer:** (A)** — `391/455` **Detailed Step-by-Step Solution & Mathematical Derivation:** P(all 4 different suits) = (4×4×4×4)/C(16,4) = 256/1820 = 64/455. P(at least 2 same suit) = 1−64/455 = 391/455.

Problem 45 • Infosys SE Question 45

CORE SE

ID: Q45 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q20)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
A shopkeeper runs a "Buy 3, Get 1 Free" offer on an item marked at ₹200 (costing him ₹120 each). A customer buys 4 items under this offer, paying for 3. What is the shopkeeper's profit percentage, based on his cost for all 4 items given?

A 25%	**B** 20%
C 30%	**D** 15%

✓ **Verified Answer:** (A)** — `25%` **Detailed Step-by-Step Solution & Mathematical Derivation:** Cost for 4 items = 4×120 = ₹480. Revenue (customer pays for 3 at ₹200) = ₹600. Profit = ₹120 on a cost of ₹480, giving 120/480×100 = 25%.

Problem 46 • Infosys SE Question 46

CORE SE

ID: Q46 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q21)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
A, B and C can individually complete a work in 15, 20 and 30 days. They work together, and the total payment of ₹9,000 is distributed in proportion to each person's work contribution. How much should C be paid?

A ₹2,000	**B** ₹1,800
C ₹2,200	**D** ₹2,500

✓ **Verified Answer:** (A)** — `₹2,000` **Detailed Step-by-Step Solution & Mathematical Derivation:** Rates in the ratio 1/15:1/20:1/30 = 4:3:2 (over LCM 60). C's share = (2/9)×9000 = ₹2,000.

Problem 47 • Infosys SE Question 47

CORE SE

ID: Q47 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q22)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
Two runners start from the same point on a 600 m circular track at the same time, in opposite directions, at 8 m/s and 10 m/s. After how many seconds will they both simultaneously be back at the starting point for the first time?

A 300 s	**B** 200 s
C 150 s	**D** 600 s

✓ **Verified Answer:** (A)** — `300 s` **Detailed Step-by-Step Solution & Mathematical Derivation:** A takes 600/8=75 s/lap; B takes 600/10=60 s/lap. Both are back at the start simultaneously at LCM(75,60) = 300 s.

Problem 48 • Infosys SE Question 48

CORE SE

ID: Q48 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q23)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
A container has 80 litres of pure milk. 20 litres of the mixture is removed and replaced with water; this process is repeated once more. How much pure milk remains?

A 45 litres	**B** 50 litres
C 40 litres	**D** 48 litres

✓ **Verified Answer:** (A)** — `45 litres` **Detailed Step-by-Step Solution & Mathematical Derivation:** Using the repeated-dilution formula: Final = $80\times(1-20/80)^2 = 80\times(3/4)^2 = 80\times9/16 = 45$ litres.

Problem 49 • Infosys SE Question 49

CORE SE

ID: Q49 | Source: Infosys PYQ | Level: Infosys SE Standard

(Infosys System Engineer ITM Mock Paper 5 - Q24)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
At a certain rate of annual compound interest, a sum doubles itself in 5 years. In how many years will it become 8 times itself, at the same rate?

A 15 years	**B** 20 years
C 10 years	**D** 25 years

✓ **Verified Answer:** (A)** — `15 years` **Detailed Step-by-Step Solution & Mathematical Derivation:** $(1+r)^5=2$ (doubling). For 8×: $(1+r)^n=8=2^3=[(1+r)^5]^3=(1+r)^{15}$, so n=15 years.